

EXPERIMENT – 2

Title:

Use var, let, const, and demonstrate basic ES6 features like Template Literals and Object Destructuring by building a Billing Calculator with user input.

Software/Tools Required:

Visual Studio Code (or any text editor), Web Browser (Google Chrome, Microsoft Edge, Firefox, etc.)

Theory:

JavaScript provides different ways to declare variables:

1. var

- var is function-scoped.
- It can be redeclared and updated.
- It is mainly used in older versions of JavaScript.

2. let

- let is block-scoped.
- It can be updated but cannot be redeclared within the same block.
- It is used for variables whose values may change during program execution.

3. const

- const is block-scoped.
- Its value cannot be reassigned after initialization.
- It is commonly used for fixed values such as constants.

4. Template Literals

- Template literals are enclosed within backticks (```).
- They allow embedding variables and expressions using `${}`.

- They make string formatting simpler and more readable.

5. Object Destructuring

- Object destructuring extracts properties from an object into separate variables.
- It improves code readability and reduces repetitive code.

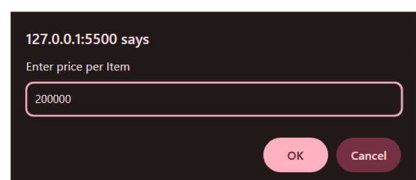
Billing Calculator

In this experiment, the Billing Calculator accepts customer details, product details, quantity, and price using `prompt()`. The total amount is calculated, GST at **18%** is added, and the final bill is displayed. The program demonstrates the practical use of `var`, `let`, `const`, template literals, and object destructuring while generating a billing receipt.

Result/Conclusion:

The experiment was successfully completed. The use of `var`, `let`, and `const` was understood and implemented effectively. ES6 features such as **Template Literals** and **Object Destructuring** were used to simplify the code and improve readability. A Billing Calculator was developed that accepts user input, calculates the total amount with **18% GST**, and displays a formatted billing receipt successfully.

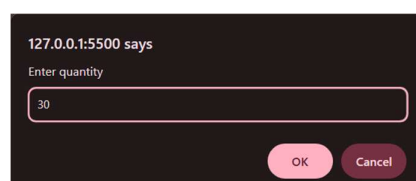
Output:



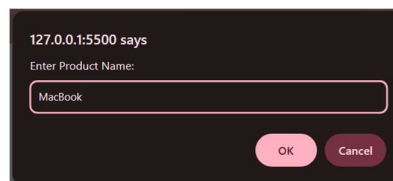
127.0.0.1:5500 says
Enter price per Item
200000
OK Cancel



127.0.0.1:5500 says
Enter Customer Name:
Radhika Chaudhari
OK Cancel



127.0.0.1:5500 says
Enter quantity
30
OK Cancel



127.0.0.1:5500 says
Enter Product Name:
MacBook
OK Cancel

Billing Calculator

Billing Receipt

Customer: Radhika Chaudhari

Product: MacBook

Quantity: 30

Price per Item: ₹200000

Total: ₹6000000

GST (18%): ₹1080000.00

Final Bill: ₹7080000.00